

SPECTRAL GARDEN — CRESCENTQ

Ethereal Phase Equalizer • 24-Band Precision • Linear/Minimum Phase Modes
Built with JUCE 8+ • VST3 / AU / Standalone

CrescentQ is a high-precision 24-band parametric equalizer designed for mixing, mastering, and surgical spectral shaping. It features switchable minimum-phase and linear-phase architectures, per-band Mid/Side routing, multi-stage oversampling, and real-time FFT visualization. The engine is optimized for low-latency operation while offering advanced phase-coherent processing for critical audio work.

This document provides a complete overview of CrescentQ's architecture, controls, and build environment.

1. CORE ARCHITECTURE & FEATURES

[1] 24-Band Parametric Array

- Up to 24 fully independent filter bands
- Filter types:
 - Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch, Bandpass
- Per-band routing:
 - Stereo, Left, Right, Mid, Side
- Logarithmic frequency mapping (20 Hz → 20 kHz)

[2] Phase & Oversampling Engine

- Minimum-phase and linear-phase modes
- Linear-phase mode includes automatic latency reporting (1024 samples)
- Oversampling tiers:
 - 1x, 2x, 4x, 8x
- Eco Mode:
 - Reduces CPU load during large sessions

[3] Visualization & Analysis

- Dual FFT analyzers (input/output)
- Real-time EQ curve rendering
- Stereo peak meters
- Internal CPU performance monitor

[4] Workflow & Presets

- A/B comparison banks (stateA / stateB)
- 25 factory presets across mixing, mastering, vocals, surgical, creative
- User preset system:
JSON-backed `.coven` files stored in local application directories

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2. PARAMETER CONTROL REFERENCE

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Global Controls

- `global_bypass` — Boolean
- `global_gain` — -30 dB → +30 dB
- `linear_phase` — Boolean
- `oversampling` — 1x / 2x / 4x / 8x
- `eco_mode` — Boolean

Band Controls (0–23)

- `band_X_active` — Boolean
- `band_X_type` — Filter topology
- `band_X_frequency` — 20 Hz → 20 kHz
- `band_X_gain` — -30 dB → +30 dB
- `band_X_q` — 0.10 → 36.00
- `band_X_mode` — Stereo / L / R / Mid / Side

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3. SYSTEM REQUIREMENTS

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Framework:

JUCE 8+

Formats:

VST3, AU, Standalone

Supported OS:

Windows, macOS

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4. BUILDING FROM SOURCE

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1. Place CrescentQ source files inside a JUCE project directory.
2. Open the `.juicer` project using Projucer.
3. Configure exporter targets (Xcode / Visual Studio).
4. Build in Release mode for optimal FFT and oversampling performance.

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